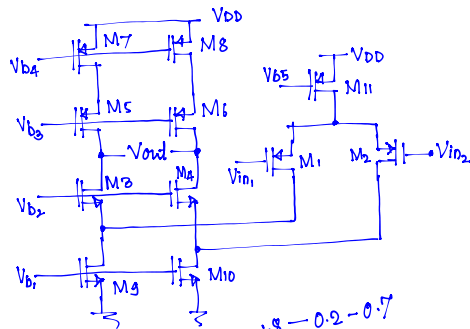




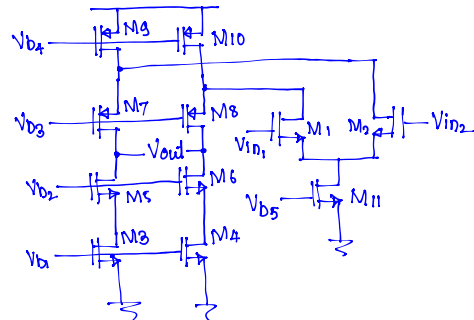
$V_{in,CM}$ range

$$V_{in,CM,max} = V_{DD} - |V_{OV11}| - |V_{GS1}|$$

$$V_{in,CM,min} = V_{OV9} - |V_{th1}|$$

$$= V_{OV9} + |V_{OV1}| - |V_{GS1}|$$

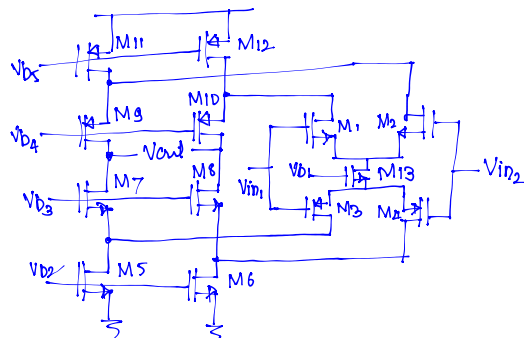
$$= V_{OV9} + |V_{OV1}| - (|V_{OV1}| + |V_{th1}|)$$



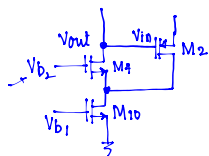
$$A_v = g_{m1,2} [g_{m8} r_{o8} (r_{o10} || r_{o2}) || g_{m6} r_{o6} r_{o4}]$$

$$V_{in,CM,min} = V_{OV11} + V_{GS1}$$

$$V_{in,CM,max} = V_{DD} - |V_{OV10}| + V_{th1}$$



If we short output with one of the input



For M4 to be in saturation

$$V_{DS4} \geq V_{GS4} - V_{th4}$$

$$V_{out} - V_{S4} \geq V_{b2} - V_{S4} - V_{th4}$$

$$V_{out} \geq V_{b2} - V_{th4}$$

For M2 to be in saturation.

$$V_{DS2} \leq V_{GS2} - V_{th2}$$

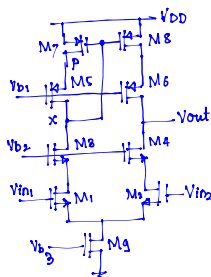
$$V_{b2} - V_{GS4} - V_{S2} \leq V_{out} - V_{S2} - V_{th2}$$

$$V_{b2}$$

$$V_{out} \geq V_{b2} - V_{GS4} + V_{th2}$$

$$V_{out} \geq V_{b2} - V_{GS4} - |V_{th2}|$$

$$0.8 - 0.6 - 0.5$$



$$V_{b1} = V_{DD} - |V_{GS7}| - |V_{th5}|$$

$$V_{pmax} = V_{DD} - |V_{OV7}|$$

$$= V_{DD} - |V_{GS7}| - |V_{th7}|$$

Gain Boosting :-

$$A_v = G_m R_{out}$$